054. Satellite Image Land Cover Classification Using Deep Learning

Area of Research: Remote Sensing, Computer Vision, and Deep Learning

Problem Statement: Accurate land cover classification is crucial for environmental monitoring, urban planning, agricultural analysis, and disaster management. Traditional classification approaches depend on manual or semi-automated techniques, which can be time-consuming, labour-intensive, and prone to errors due to differences in satellite imagery caused by seasonal changes, atmospheric conditions, and different sensor resolutions.

This project aims to develop a robust deep learning-based system for automated land cover classification from satellite images. By leveraging state-of-the-art deep learning architectures, the model will effectively classify various land cover types, such as water bodies, vegetation, built-up areas, and barren land. The goal is to achieve high classification accuracy and generalization across diverse geographic regions, enabling efficient large-scale land cover mapping for sustainable development and decision-making.

Dataset: The dataset contains high-resolution satellite images with labelled land cover categories, providing a rich source of data for training and evaluating deep learning models. It includes annotations for multiple land cover classes, ensuring a comprehensive benchmark for classification tasks.

Dataset:

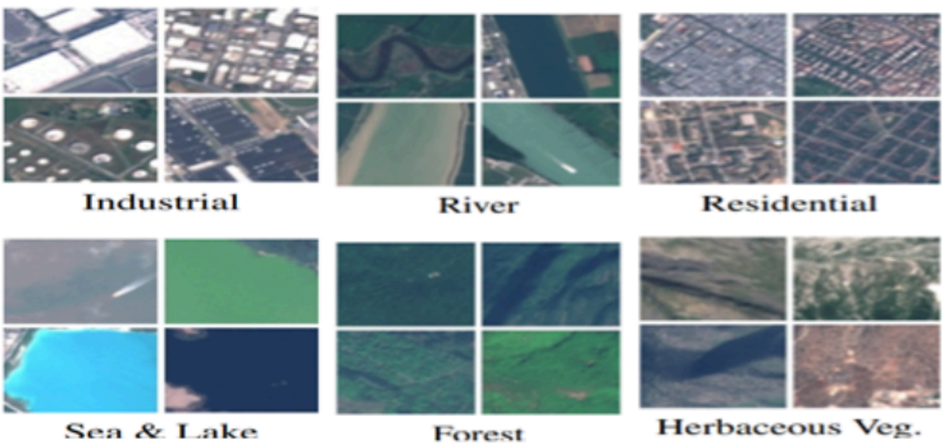

Figure 54: Example of a satellite image with different land cover types.

Dataset URL: https://zenodo.org/records/7711810

Task: Develop a deep learning model capable of classifying land cover types in satellite images with high accuracy. The model should be robust to variations in lighting, resolution, and seasonal effects, ensuring generalizability across different geographic regions.

Relevant Papers:

[1] Helber, Patrick, et al. "Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification." IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 12.7 (2019): 2217-2226.

[2] Helber, Patrick, et al. "Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification." IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 12.7 (2019): 2217-2226.

[3] Bischke, Benjamin, et al. "Detection of Flooding Events in Social Multimedia and Satellite Imagery using Deep Neural Networks." MediaEval. 2017.

[4] Zhao, Shengyu, et al. "Land use and land cover classification meets deep learning: a review." Sensors 23.21 (2023): 8966.